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1. Project Description

The XYZ organization aims to develop a comprehensive **Airline Management System** using a Database Management System to efficiently manage its operational, administrative, and customer service processes. The system is designed to handle all core airline activities including employee management, flight scheduling, passenger bookings, ticketing, baggage handling, maintenance operations, and financial transactions.

The system maintains detailed records of employees, their roles, and their assignments to flights and maintenance events. Aircraft information is stored along with seating configurations and maintenance history. Flights are defined through routes and are scheduled using flight schedules, which capture real-time operational data such as departure and arrival timings.

Passengers can make bookings, receive tickets for specific flights, and select or be assigned seats. The system supports multi-leg journeys through connections and connection segments, allowing passengers to travel across multiple flights under a single itinerary. Check-in and boarding processes are recorded to track passenger movement through the airport.

Baggage management is integrated with check-in to ensure accurate tracking of passenger luggage. Payment and invoice modules handle all financial transactions, ensuring traceability and accountability. The system also supports reporting capabilities for management, enabling analysis of flight operations, revenue, passenger trends, and resource utilization.

This system ensures data consistency, eliminates redundancy through normalization, and supports real-world airline operational complexity with scalable and structured database design.

2. Detailed Analysis using Nouns and Verbs

The XYZ organization develops a **Database Management System** for an **Airline Management System** that manages **employees**, **flights**, **passengers**, **bookings**, **tickets**, **baggage**, **payments**, and **aircraft operations**.

The system **stores** and **manages employee records**, including their **roles**, **assignments**, and **maintenance activities**. It also **tracks aircraft**, their **models**, **seating configurations**, and **maintenance events**.

The system **defines flights** using **routes** and **schedules**, and **assigns aircraft** and **crew members** to each **flight schedule**. **Passengers** can **make bookings**, **receive tickets**, and **select seats** for specific **flights**.

The system **handles multi-leg journeys** using **connections** and **connection segments**, allowing **passengers** to **travel** across multiple **flights**. It also **records check-in** and **boarding events** to **track** passenger movement.

The system **manages baggage**, which is **linked** to **check-in records**, ensuring accurate **tracking** and **handling**. It also **processes payments** and **generates invoices** for each **booking**, maintaining financial **records**.

Finally, the system **generates reports** for **management**, including **flight performance**, **revenue analysis**, **passenger statistics**, and **resource utilization**.

Nouns

Employee, Role, Aircraft, Flight, Route, Airport, Passenger, Booking, Ticket, Seat, Flight Schedule, Connection, Check-In, Boarding, Baggage, Payment, Invoice, Maintenance, Database, System, Report, Management

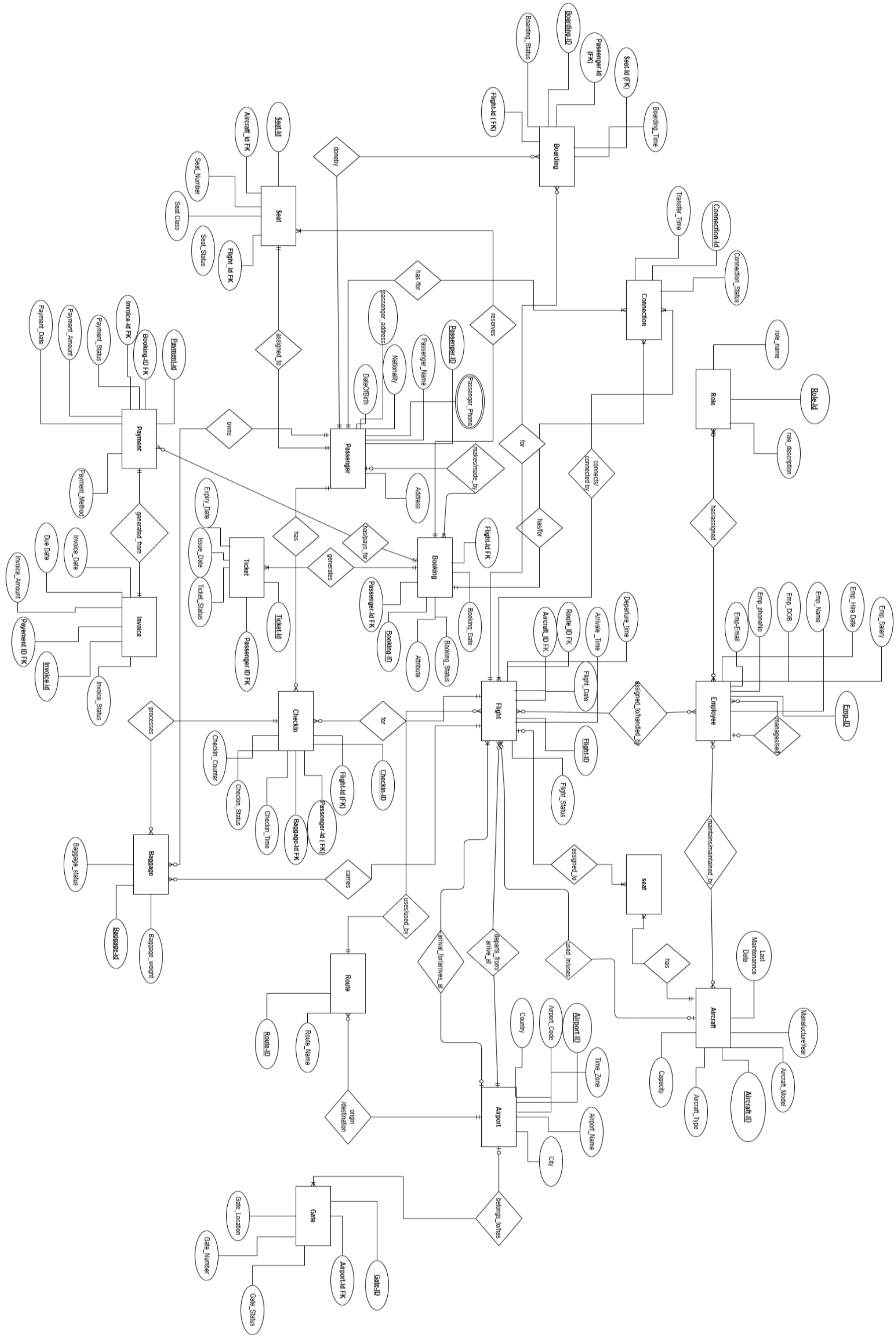
Verbs

Store, Manage, Track, Assign, Define, Make, Receive, Select, Handle, Record, Process, Generate, Travel, Link

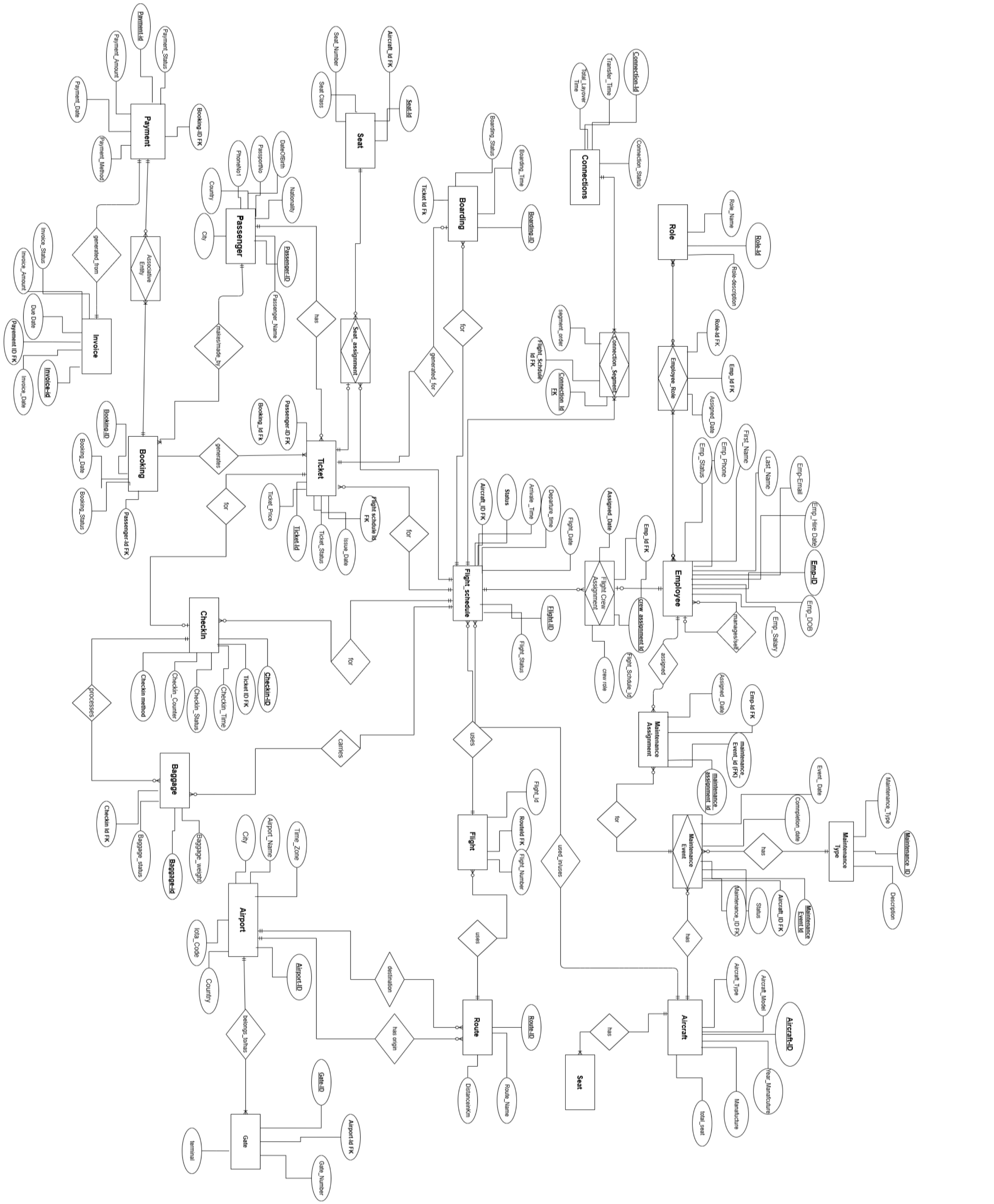
Adjectives

Comprehensive, Operational, Financial, Multi-leg, Accurate, Real-time, Structured, Normalized, Scalable

3. Entity Relationship Diagram (ERD) :



4. Normalized Entity Relationship Diagram



5. Relational Model

1. Employee

employee_id	manager_id	first_name	last_name	email	phone	date_of_birth	hire_date	employment_status
INT	INT	VARCHAR(50)	VARCHAR(50)	VARCHAR(100)	VARCHAR(20)	DATE	DATE	VARCHAR(20)
PK, NOT NULL	FK → Employee(employee_id), NULLABLE	NOT NULL	NOT NULL	NOT NULL, UNIQUE			NOT NULL	NOT NULL, DEFAULT 'Active'

2. Role

role_id	role_name	role_description
INT	VARCHAR(100)	VARCHAR(255)
PK, NOT NULL	NOT NULL, UNIQUE	

3. EmployeeRole *(Associative)*

employee_role_id	employee_id	role_id	assigned_date
INT	INT	INT	DATE
PK, NOT NULL	FK → Employee(employee_id), NOT NULL	FK → Role(role_id), NOT NULL	NOT NULL

4. MaintenanceType

maintenance_type_id	type_name	type_description
INT	VARCHAR(100)	VARCHAR(255)
PK, NOT NULL	NOT NULL, UNIQUE	

5. MaintenanceEvent

maintenance_event_id	maintenance_type_id	event_date	completion_date	event_description	status
INT	INT	DATE	DATE	VARCHAR(255)	VARCHAR(30)
PK, NOT NULL	FK → MaintenanceType(maintenance_type_id), NOT NULL	NOT NULL	NULLABLE		NOT NULL

---

6. MaintenanceAssignment *(Associative)*

maintenance_assignment_id	employee_id	maintenance_event_id	assigned_date
INT	INT	INT	DATE
PK, NOT NULL	FK → Employee(employee_id), NOT NULL	FK → MaintenanceEvent(maintenance_event_id), NOT NULL	NOT NULL

---

7. Airport

airport_id	airport_name	iata_code	city	country
INT	VARCHAR(150)	CHAR(3)	VARCHAR(100)	VARCHAR(100)
PK, NOT NULL	NOT NULL	NOT NULL, UNIQUE	NOT NULL	NOT NULL

---

8. Gate

gate_id	airport_id	gate_number	terminal
INT	INT	VARCHAR(10)	VARCHAR(50)
PK, NOT NULL	FK → Airport(airport_id), NOT NULL	NOT NULL	NULLABLE

9. Route

route_id	origin_airport_id	destination_airport_id	distance_km	estimated_duration_min
INT	INT	INT	DECIMAL(8,2)	INT
PK, NOT NULL	FK → Airport(airport_id), NOT NULL	FK → Airport(airport_id), NOT NULL	NULLABLE	NULLABLE

---

10. Flight

flight_id	route_id	flight_number	airline_name	base_price
INT	INT	VARCHAR(10)	VARCHAR(100)	DECIMAL(10,2)
PK, NOT NULL	FK → Route(route_id), NOT NULL	NOT NULL, UNIQUE	NOT NULL	NOT NULL

---

11. Aircraft

aircraft_id	model	manufacturer	registration_number	total_seats	year_manufactured
INT	VARCHAR(100)	VARCHAR(100)	VARCHAR(20)	INT	YEAR
PK, NOT NULL	NOT NULL	NOT NULL	NOT NULL, UNIQUE	NOT NULL	NULLABLE

---

12. Seat

seat_id	aircraft_id	seat_number	seat_class
INT	INT	VARCHAR(10)	VARCHAR(20)
PK, NOT NULL	FK → Aircraft(aircraft_id), NOT NULL	NOT NULL	NOT NULL

13. FlightSchedule

flight_schedule_id	flight_id	aircraft_id	scheduled_departure	scheduled_arrival	actual_departure	actual_arrival	status
INT	INT	INT	DATETIME	DATETIME	DATETIME	DATETIME	VARCHAR(30)
PK, NOT NULL	FK → Flight(flight_id), NOT NULL	FK → Aircraft(aircraft_id), NOT NULL	NOT NULL	NOT NULL	NULLABLE	NULLABLE	NOT NULL, DEFAULT 'Scheduled'

14. FlightCrewAssignment *(Associative)*

flight_crew_assignment_id	employee_id	flight_schedule_id	crew_role	assigned_date

INT	INT	INT	VARCHAR (50)	DATE
PK, NOT NULL	FK → Employee(employee_id), NOT NULL	FK → FlightSchedule(flight_schedule_id), NOT NULL	NULLABLE	NOT NULL

15. Connection

connection_id	connection_name	connection_type	total_layover_duration_min
INT	VARCHAR(100)	VARCHAR(50)	INT
PK, NOT NULL	NOT NULL	NULLABLE	NULLABLE

---

16. ConnectionSegment *(Associative)*

connection_segment_id	connection_id	flight_schedule_id	segment_order
INT	INT	INT	INT
PK, NOT NULL	FK → Connection(connection_id), NOT NULL	FK → FlightSchedule(flight_schedule_id), NOT NULL	NOT NULL

Table-Level Constraint: UNIQUE(connection_id, flight_schedule_id)

17. Passenger

passenger_id	first_name	last_name	passport_number	nationality	date_of_birth	email	phone
INT	VARCHAR(50)	VARCHAR(50)	VARCHAR(30)	VARCHAR(50)	DATE	VARCHAR (100)	VARCHAR(20)
PK, NOT NULL	NOT NULL	NOT NULL	NOT NULL, UNIQUE	NOT NULL	NOT NULL	NOT NULL, UNIQUE	NULLABLE

18. Booking

booking_id	passenger_id	booking_date	total_amount	booking_status
INT	INT	DATETIME	DECIMAL(10,2)	VARCHAR(30)
PK, NOT NULL	FK → Passenger(passenger_id), NOT NULL	NOT NULL	NOT NULL	NOT NULL, DEFAULT 'Confirmed'

19. Ticket

ticket_id	booking_id	passenger_id	flight_schedule_id	ticket_number	ticket_price	class_type	ticket_status
INT	INT	INT	INT	VARCHAR(20)	DECIMAL(10,2)	VARCHAR(20)	VARCHAR(30)
PK, NOT NULL	FK → Booking(booking_id), NOT NULL	FK → Passenger(passenger_id), NOT NULL	FK → FlightSchedule(flight_schedule_id), NOT NULL	NOT NULL, UNIQUE	NOT NULL	NOT NULL	NOT NULL, DEFAULT 'Active'

20. SeatAssignment *(Associative)*

seat_assignment_id	seat_id	ticket_id	flight_schedule_id	assignment_datetime
INT	INT	INT	INT	DATETIME
PK, NOT NULL	FK → Seat(seat_id), NOT NULL	FK → Ticket(ticket_id), NULLABLE	FK → FlightSchedule(flight_schedule_id), NOT NULL	NULLABLE

21. CheckIn

check_in_id	ticket_id	check_in_datetime	check_in_method	counter_number
INT	INT	DATETIME	VARCHAR(30)	VARCHAR(10)
PK, NOT NULL	FK → Ticket(ticket_id), NOT NULL	NOT NULL	NULLABLE	NULLABLE

22. Boarding

boarding_id	ticket_id	boarding_datetime	boarding_status
INT	INT	DATETIME	VARCHAR(30)
PK, NOT NULL	FK → Ticket(ticket_id), NOT NULL	NOT NULL	NOT NULL

23. Baggage

baggage_id	ticket_id	weight_kg	baggage_type	baggage_tag_number	baggage_status
INT	INT	DECIMAL(5,2)	VARCHAR(30)	VARCHAR(30)	VARCHAR(20)
PK, NOT NULL	FK → Ticket(ticket_id), NOT NULL	NOT NULL	NULLABLE	UNIQUE, NULLABLE	NOT NULL

24. Payment

payment_id	booking_id	payment_datetime	payment_amount	payment_method	payment_status	transaction_reference
INT	INT	DATETIME	DECIMAL(10,2)	VARCHAR(50)	VARCHAR(30)	VARCHAR(100)
PK, NOT NULL	FK → Booking(booking_id), NOT NULL	NOT NULL	NOT NULL	NOT NULL	NOT NULL	UNIQUE, NULLABLE

25. Invoice

invoice_id	payment_id	invoice_date	invoice_amount	billing_address	invoice_status
INT	INT	DATE	DECIMAL(10,2)	VARCHAR(255)	VARCHAR(30)
PK, NOT NULL	FK → Payment(payment_id), NOT NULL, UNIQUE	NOT NULL	NOT NULL	NULLABLE	NOT NULL

6. SQL Queries

- CREATE TABLE statements

```sql
CREATE TABLE Employee (
   employee_id INT PRIMARY KEY,
   manager_id INT NULL,
   first_name VARCHAR(50) NOT NULL,
   last_name VARCHAR(50) NOT NULL,
   email VARCHAR(100) NOT NULL UNIQUE,
   phone VARCHAR(20),
   date_of_birth DATE NOT NULL,
   hire_date DATE NOT NULL,
   employment_status VARCHAR(20) NOT NULL DEFAULT 'Active',

   FOREIGN KEY (manager_id) REFERENCES Employee(employee_id)
);

CREATE TABLE Role (
   role_id INT PRIMARY KEY,
   role_name VARCHAR(100) NOT NULL UNIQUE,
   role_description VARCHAR(255)
);

CREATE TABLE EmployeeRole (
   employee_role_id INT PRIMARY KEY,
   employee_id INT NOT NULL,
   role_id INT NOT NULL,
   assigned_date DATE NOT NULL,

   UNIQUE (employee_id, role_id),

   FOREIGN KEY (employee_id) REFERENCES Employee(employee_id),
   FOREIGN KEY (role_id) REFERENCES Role(role_id)
);

CREATE TABLE Airport (
   airport_id INT PRIMARY KEY,
   airport_name VARCHAR(150) NOT NULL,
   iata_code CHAR(3) NOT NULL UNIQUE,
   city VARCHAR(100) NOT NULL,
   country VARCHAR(100) NOT NULL
);

CREATE TABLE Gate (
   gate_id INT PRIMARY KEY,
   airport_id INT NOT NULL,
   gate_number VARCHAR(10) NOT NULL,
   terminal VARCHAR(50),

   UNIQUE (airport_id, gate_number),

   FOREIGN KEY (airport_id) REFERENCES Airport(airport_id)
);
```

```sql
CREATE TABLE Route (
    route_id INT PRIMARY KEY,
    origin_airport_id INT NOT NULL,
    destination_airport_id INT NOT NULL,
    distance_km DECIMAL(8,2),
    estimated_duration_min INT,

    FOREIGN KEY (origin_airport_id) REFERENCES Airport(airport_id),
    FOREIGN KEY (destination_airport_id) REFERENCES Airport(airport_id),

    CHECK (origin_airport_id <> destination_airport_id)
);

CREATE TABLE Flight (
    flight_id INT PRIMARY KEY,
    route_id INT NOT NULL,
    flight_number VARCHAR(10) NOT NULL UNIQUE,
    airline_name VARCHAR(100) NOT NULL,
    base_price DECIMAL(10,2) NOT NULL,

    FOREIGN KEY (route_id) REFERENCES Route(route_id)
);

CREATE TABLE Aircraft (
    aircraft_id INT PRIMARY KEY,
    model VARCHAR(100) NOT NULL,
    manufacturer VARCHAR(100) NOT NULL,
    registration_number VARCHAR(20) NOT NULL UNIQUE,
    total_seats INT NOT NULL,
    year_manufactured SMALLINT
);

CREATE TABLE Seat (
    seat_id INT PRIMARY KEY,
    aircraft_id INT NOT NULL,
    seat_number VARCHAR(10) NOT NULL,
    seat_class VARCHAR(20) NOT NULL,

    UNIQUE (aircraft_id, seat_number),

    FOREIGN KEY (aircraft_id) REFERENCES Aircraft(aircraft_id)
);

CREATE TABLE FlightSchedule (
    flight_schedule_id INT PRIMARY KEY,
    flight_id INT NOT NULL,
    aircraft_id INT NOT NULL,
    scheduled_departure DATETIME NOT NULL,
    scheduled_arrival DATETIME NOT NULL,
    actual_departure DATETIME,
    actual_arrival DATETIME,
    status VARCHAR(30) NOT NULL DEFAULT 'Scheduled',

    FOREIGN KEY (flight_id) REFERENCES Flight(flight_id),
```

```sql
    FOREIGN KEY (aircraft_id) REFERENCES Aircraft(aircraft_id),


    CHECK (scheduled_arrival > scheduled_departure)
);

CREATE TABLE Passenger (
    passenger_id INT PRIMARY KEY,
    first_name VARCHAR(50) NOT NULL,
    last_name VARCHAR(50) NOT NULL,
    passport_number VARCHAR(30) NOT NULL UNIQUE,
    nationality VARCHAR(50) NOT NULL,
    date_of_birth DATE NOT NULL,
    email VARCHAR(100) NOT NULL UNIQUE,
    phone VARCHAR(20)
);

CREATE TABLE Booking (
    booking_id INT PRIMARY KEY,
    booking_date DATETIME NOT NULL,
    total_amount DECIMAL(10,2) NOT NULL,
    booking_status VARCHAR(30) NOT NULL DEFAULT 'Confirmed'
);

CREATE TABLE Ticket (
    ticket_id INT PRIMARY KEY,
    booking_id INT NOT NULL,
    passenger_id INT NOT NULL,
    flight_schedule_id INT NOT NULL,
    ticket_number VARCHAR(20) NOT NULL UNIQUE,
    ticket_price DECIMAL(10,2) NOT NULL,
    class_type VARCHAR(20) NOT NULL,
    ticket_status VARCHAR(30) NOT NULL DEFAULT 'Active',

    FOREIGN KEY (booking_id) REFERENCES Booking(booking_id),
    FOREIGN KEY (passenger_id) REFERENCES Passenger(passenger_id),
    FOREIGN KEY (flight_schedule_id) REFERENCES FlightSchedule(flight_schedule_id)
);

CREATE TABLE CheckIn (
    check_in_id INT PRIMARY KEY,
    ticket_id INT NOT NULL UNIQUE ,
    check_in_datetime DATETIME NOT NULL,
    check_in_method VARCHAR(30),
    counter_number VARCHAR(10),

   FOREIGN KEY (ticket_id)
   REFERENCES Ticket(ticket_id)
);

CREATE TABLE Boarding (
    boarding_id INT PRIMARY KEY,
    ticket_id INT NOT NULL,
    boarding_datetime DATETIME NOT NULL,
    boarding_status VARCHAR(30) NOT NULL,
```

```sql
    FOREIGN KEY (ticket_id)
    REFERENCES CheckIn(ticket_id)
);

CREATE TABLE Connections (
    connection_id INT PRIMARY KEY,
    connection_name VARCHAR(100) NOT NULL,
    connection_type VARCHAR(50),
    total_layover_duration_min INT
);

CREATE TABLE ConnectionSegment (
    connection_segment_id INT PRIMARY KEY,
    connection_id INT NOT NULL,
    flight_schedule_id INT NOT NULL,
    segment_order INT NOT NULL,

    UNIQUE (connection_id, flight_schedule_id),

    FOREIGN KEY (connection_id) REFERENCES Connections(connection_id),
    FOREIGN KEY (flight_schedule_id) REFERENCES FlightSchedule(flight_schedule_id),

    CHECK (segment_order > 0)
);

CREATE TABLE MaintenanceType (
    maintenance_type_id INT PRIMARY KEY,
    type_name VARCHAR(100) NOT NULL UNIQUE,
    type_description VARCHAR(255)
);

CREATE TABLE MaintenanceEvent (
    maintenance_event_id INT PRIMARY KEY,
    aircraft_id INT NOT NULL,
    maintenance_type_id INT NOT NULL,
    event_date DATE NOT NULL,
    completion_date DATE,
    event_description VARCHAR(255),
    status VARCHAR(30) NOT NULL,

    FOREIGN KEY (maintenance_type_id) REFERENCES MaintenanceType(maintenance_type_id),
    FOREIGN KEY (aircraft_id) REFERENCES Aircraft(aircraft_id),
    CHECK (completion_date IS NULL OR completion_date >= event_date)
);

CREATE TABLE MaintenanceAssignment (
    maintenance_assignment_id INT PRIMARY KEY,
    employee_id INT NOT NULL,
    maintenance_event_id INT NOT NULL,
    assigned_date DATE NOT NULL,

    UNIQUE (employee_id, maintenance_event_id),

    FOREIGN KEY (employee_id) REFERENCES Employee(employee_id),
```

```sql
    FOREIGN KEY (maintenance_event_id) REFERENCES MaintenanceEvent(maintenance_event_id)
);



CREATE TABLE FlightCrewAssignment (
    flight_crew_assignment_id INT PRIMARY KEY,
    employee_id INT NOT NULL,
    flight_schedule_id INT NOT NULL,
    crew_role VARCHAR(50),
    assigned_date DATE NOT NULL,

    UNIQUE (employee_id, flight_schedule_id),

    FOREIGN KEY (employee_id) REFERENCES Employee(employee_id),
    FOREIGN KEY (flight_schedule_id) REFERENCES FlightSchedule(flight_schedule_id)
);

CREATE TABLE SeatAssignment (
    seat_assignment_id INT PRIMARY KEY,
    seat_id INT NOT NULL,
    ticket_id INT NOT NULL,
    flight_schedule_id INT NOT NULL,
    assignment_datetime DATETIME,

    UNIQUE (seat_id, flight_schedule_id),

    FOREIGN KEY (seat_id) REFERENCES Seat(seat_id),
    FOREIGN KEY (ticket_id) REFERENCES Ticket(ticket_id),
    FOREIGN KEY (flight_schedule_id) REFERENCES FlightSchedule(flight_schedule_id)
);

CREATE TABLE Baggage (
    baggage_id INT PRIMARY KEY,
    check_in_id INT NOT NULL,
    weight_kg DECIMAL(5,2) NOT NULL,
    baggage_type VARCHAR(30),
    baggage_tag_number VARCHAR(30) UNIQUE,
    baggage_status VARCHAR(20) NOT NULL,


    FOREIGN KEY (check_in_id) REFERENCES CheckIn(check_in_id),


    CHECK (weight_kg > 0)
);

CREATE TABLE Payment (
    payment_id INT PRIMARY KEY,
    booking_id INT NOT NULL,
    payment_datetime DATETIME NOT NULL,
    payment_amount DECIMAL(10,2) NOT NULL,
    payment_method VARCHAR(50) NOT NULL,
    payment_status VARCHAR(30) NOT NULL,
    transaction_reference VARCHAR(100) UNIQUE,
```

```sql
    FOREIGN KEY (booking_id) REFERENCES Booking(booking_id),

    CHECK (payment_amount > 0)
);

CREATE TABLE Invoice (
    invoice_id INT PRIMARY KEY,
    payment_id INT NOT NULL UNIQUE,
    invoice_date DATE NOT NULL,
    invoice_amount DECIMAL(10,2) NOT NULL,
    billing_address VARCHAR(255),
    invoice_status VARCHAR(30) NOT NULL,

    FOREIGN KEY (payment_id) REFERENCES Payment(payment_id),

    CHECK (invoice_amount > 0)
);

CREATE TABLE BookingPassenger (
    booking_id INT,
    passenger_id INT,

    PRIMARY KEY (booking_id, passenger_id),

    FOREIGN KEY (booking_id) REFERENCES Booking(booking_id),
    FOREIGN KEY (passenger_id) REFERENCES Passenger(passenger_id)
);
```

- Alter Table

```sql
ALTER TABLE Aircraft
ADD is_active BIT NOT NULL DEFAULT 1;
```

Populate Data into tables as:

```sql
INSERT INTO Role (role_id, role_name, role_description) VALUES
(1, 'Captain', 'Pilot in command of the aircraft'),
(2, 'First Officer', 'Co-pilot of the aircraft'),
(3, 'Lead Cabin Crew', 'In-charge of flight attendants'),
(4, 'Flight Attendant', 'Cabin crew member'),
(5, 'Dispatcher', 'Flight planning and dispatch operations'),
(19, 'Ground Equipment Mechanic', 'Maintains airport support vehicles'),
(20, 'Station Manager', 'Overall airport station operations');

INSERT INTO Employee (employee_id, manager_id, first_name, last_name, email, phone, date_of_birth,
hire_date, employment_status) VALUES
(1, NULL, 'Tariq', 'Mehmood', 'tariq.m@airline.com', '0300-1111111', '1975-04-12', '2005-01-15', 'Active'),
(2, 1, 'Ali', 'Khan', 'ali.k@airline.com', '0300-2222222', '1980-08-22', '2010-03-01', 'Active'),
(3, 1, 'Ayesha', 'Siddiqa', 'ayesha.s@airline.com', '0300-3333333', '1982-11-05', '2012-06-15', 'Active'),
(4, 2, 'Bilal', 'Ahmed', 'bilal.a@airline.com', '0300-4444444', '1990-02-14', '2015-09-20', 'Active'),
(5, 2, 'Fatima', 'Noor', 'fatima.n@airline.com', '0300-5555555', '1985-07-30', '2014-11-10', 'On Leave'),
```

```sql
(6, 3, 'Usman', 'Tariq', 'usman.t@airline.com', '0300-6666666', '1992-05-18', '2018-01-05', 'Active'),
(7, 3, 'Zainab', 'Ali', 'zainab.a@airline.com', '0300-7777777', '1988-09-25', '2016-04-12', 'Active'),
(8, 1, 'Hassan', 'Raza', 'hassan.r@airline.com', '0300-8888888', '1979-12-01', '2008-08-08', 'Active'),
'2018-08-08', 'Active'),
(24, 2, 'Fahad', 'Mustafa', 'fahad.m@airline.com', '0302-6666666', '1985-09-09', '2016-06-16', 'Active'),
(25, 3, 'Mahira', 'Khan', 'mahira.k@airline.com', '0302-7777777', '1988-12-21', '2019-02-02', 'Active');

INSERT INTO EmployeeRole (employee_role_id, employee_id, role_id, assigned_date) VALUES
(1, 1, 15, '2005-01-20'), (2, 2, 1, '2010-03-10'), (3, 3, 3, '2012-06-20'), (4, 4, 2, '2015-09-25'), (5, 5, 4, '2014-11-
15'),
(6, 6, 2, '2018-01-10'), (7, 7, 4, '2016-04-15'), (8, 8, 9, '2008-08-10'), (9, 9, 10, '2011-05-25'), (10, 10, 11, '2020-
02-20'),
(11, 11, 1, '2013-07-10'), (12, 12, 12, '2017-09-15'), (13, 13, 5, '2019-11-15'), (14, 14, 6, '2021-03-20'), (15, 15, 7,
'2014-08-20'),
(16, 16, 14, '2022-01-15'), (17, 17, 13, '2009-10-15'), (18, 18, 1, '2006-06-10'), (19, 19, 4, '2023-05-05'), (20, 20,
8, '2024-01-05'),
(21, 21, 1, '2004-04-10'), (22, 22, 10, '2010-10-15'), (50, 20, 13, '2024-02-01');

INSERT INTO Airport (airport_id, airport_name, iata_code, city, country) VALUES
(1, 'Jinnah International Airport', 'KHI', 'Karachi', 'Pakistan'),
(2, 'Allama Iqbal International Airport', 'LHE', 'Lahore', 'Pakistan'),
(3, 'Islamabad International Airport', 'ISB', 'Islamabad', 'Pakistan'),
(4, 'Bacha Khan International Airport', 'PEW', 'Peshawar', 'Pakistan'),
(5, 'Multan International Airport', 'MUX', 'Multan', 'Pakistan'),
(6, 'Quetta International Airport', 'UET', 'Quetta', 'Pakistan'),
(7, 'Faisalabad International Airport', 'LYP', 'Faisalabad', 'Pakistan'),
(8, 'Sialkot International Airport', 'SKT', 'Sialkot', 'Pakistan'),
(9, 'Gwadar International Airport', 'GWD', 'Gwadar', 'Pakistan'),
(19, 'Beijing Capital International Airport', 'PEK', 'Beijing', 'China'),
(20, 'Sydney Kingsford Smith Airport', 'SYD', 'Sydney', 'Australia');

INSERT INTO Gate (gate_id, airport_id, gate_number, terminal) VALUES
(1, 1, 'G1', 'Terminal 1'),
(2, 1, 'G2', 'Terminal 1'),
(3, 1, 'G3', 'Terminal 2'),
(4, 1, 'G4', 'Terminal 2'),
(5, 1, 'G5', 'Terminal M'),
(6, 2, 'G1', 'Main Terminal'),
(7, 2, 'G2', 'Main Terminal'),
(8, 2, 'G3', 'Main Terminal'),
(9, 2, 'G4', 'Hajj Terminal'),
(50, 20, 'G2', 'Terminal 1');

INSERT INTO Route (route_id, origin_airport_id, destination_airport_id, distance_km, estimated_duration_min)
VALUES
(1, 1, 2, 1020.50, 105),
(2, 2, 1, 1020.50, 105),
(3, 1, 3, 1130.00, 120),
(4, 3, 1, 1130.00, 120),
(5, 2, 3, 270.00, 45),
(6, 3, 2, 270.00, 45),
(7, 1, 4, 1090.00, 115),
(8, 4, 1, 1090.00, 115),
(9, 2, 4, 380.00, 60),
(10, 4, 2, 380.00, 60),
```

```sql
(50, 20, 1, 10980.00, 800);

INSERT INTO Flight (flight_id, route_id, flight_number, airline_name, base_price) VALUES
(1, 1, 'PK302', 'Pakistan International Airlines', 15000.00),
(2, 2, 'PK303', 'Pakistan International Airlines', 15000.00),
(3, 3, 'PK300', 'Pakistan International Airlines', 18000.00),
(4, 4, 'PK301', 'Pakistan International Airlines', 18000.00),
(5, 5, 'PA401', 'Airblue', 10000.00),
(6, 6, 'PA402', 'Airblue', 10000.00),
(7, 7, 'ER501', 'SereneAir', 16000.00),
(8, 8, 'ER502', 'SereneAir', 16000.00),
(9, 9, '9P601', 'Fly Jinnah', 9000.00),
(49, 49, 'PK800', 'Pakistan International Airlines', 160000.00), (50, 50, 'PK801', 'Pakistan International Airlines',
160000.00);

INSERT INTO Aircraft (aircraft_id, model, manufacturer, registration_number, total_seats, year_manufactured,
is_active) VALUES
(1, 'A320-200', 'Airbus', 'AP-BLA', 160, 2010, 1),
(2, 'A320-200', 'Airbus', 'AP-BLB', 160, 2012, 1),
(3, 'A320-200', 'Airbus', 'AP-BLC', 160, 2014, 1),
(4, 'B777-200ER', 'Boeing', 'AP-BGJ', 320, 2005, 1),
(5, 'B777-200ER', 'Boeing', 'AP-BGK', 320, 2006, 1),
(6, 'B777-300ER', 'Boeing', 'AP-BID', 380, 2008, 1),
(7, 'A321neo', 'Airbus', 'AP-BOM', 210, 2021, 1),
(8, 'A321neo', 'Airbus', 'AP-BON', 210, 2022, 1),
(9, 'B737-800', 'Boeing', 'AP-BPA', 180, 2015, 1
(20, 'B777-200ER', 'Boeing', 'G-YMMB', 270, 2004, 1);

INSERT INTO Seat (seat_id, aircraft_id, seat_number, seat_class) VALUES
(1, 1, '1A', 'Business'),
(2, 1, '1B', 'Business'),
(3, 1, '10A', 'Economy'),
(4, 1, '10B', 'Economy'),
(5, 1, '10C', 'Economy'),
(6, 2, '1A', 'Business'),
(7, 2, '1B', 'Business'),
(8, 2, '10A', 'Economy'),
(50, 10, '11C', 'Economy');

INSERT INTO FlightSchedule (flight_schedule_id, flight_id, aircraft_id, scheduled_departure, scheduled_arrival,
actual_departure, actual_arrival, status) VALUES
(1, 1, 1, '2024-06-01 08:00:00', '2024-06-01 09:45:00', '2024-06-01 08:05:00', '2024-06-01 09:50:00',
'Completed'),
(2, 2, 2, '2024-06-01 10:30:00', '2024-06-01 12:15:00', '2024-06-01 10:30:00', '2024-06-01 12:10:00',
'Completed'),
(3, 3, 3, '2024-06-01 14:00:00', '2024-06-01 16:00:00', NULL, NULL, 'Scheduled'),
(4, 4, 1, '2024-06-02 09:00:00', '2024-06-02 11:00:00', NULL, NULL, 'Scheduled'),
(5, 5, 7, '2024-06-02 15:00:00', '2024-06-02 15:45:00', NULL, NULL, 'Scheduled'),
(6, 6, 8, '2024-06-03 07:00:00', '2024-06-03 07:45:00', NULL, NULL, 'Scheduled'),
(7, 7, 9, '2024-06-03 18:00:00', '2024-06-03 19:55:00', NULL, NULL, 'Scheduled'),
(8, 8, 10, '2024-06-04 11:00:00', '2024-06-04 12:55:00', NULL, NULL, 'Scheduled'),
(9, 9, 2, '2024-06-04 16:00:00', '2024-06-04 17:00:00', NULL, NULL, 'Scheduled'),
(50, 50, 6, '2024-06-25 09:00:00', '2024-06-25 22:20:00', NULL, NULL, 'Scheduled');
```

```sql
INSERT INTO Passenger (passenger_id, first_name, last_name, passport_number, nationality, date_of_birth,
email, phone) VALUES
(1, 'Salman', 'Khan', 'PK1111222', 'Pakistani', '1985-05-15', 'salman.k@mail.com', '0321-1111111'),
(2, 'Mahnoor', 'Baloch', 'PK2222333', 'Pakistani', '1990-08-20', 'mahnoor.b@mail.com', '0321-2222222'),
(3, 'Fawad', 'Khan', 'PK3333444', 'Pakistani', '1982-11-29', 'fawad.k@mail.com', '0321-3333333'),
(4, 'Sajal', 'Ali', 'PK4444555', 'Pakistani', '1994-01-17', 'sajal.a@mail.com', '0321-4444444'),
(5, 'Atif', 'Aslam', 'PK5555666', 'Pakistani', '1983-03-12', 'atif.a@mail.com', '0321-5555555'),
(6, 'Mehwish', 'Hayat', 'PK6666777', 'Pakistani', '1988-01-06', 'mehwish.h@mail.com', '0321-6666666'),
(7, 'Humayun', 'Saeed', 'PK7777888', 'Pakistani', '1971-05-09', 'humayun.s@mail.com', '0321-7777888'),
(8, 'Ayeza', 'Khan', 'PK8888999', 'Pakistani', '1991-01-15', 'ayeza.k@mail.com', '0321-8888999'),
(9, 'Babar', 'Azam', 'PK9999000', 'Pakistani', '1994-10-15', 'babar.a@mail.com', '0321-9999000'),
(10, 'Shaheen', 'Afridi', 'PK1010101', 'Pakistani', '2000-04-06', 'shaheen.a@mail.com', '0321-1010101');

INSERT INTO Booking (booking_id, booking_date, total_amount, booking_status) VALUES
(1, '2024-05-01 10:00:00', 15000.00, 'Confirmed'),
(2, '2024-05-02 11:00:00', 15000.00, 'Confirmed'),
(3, '2024-05-03 12:00:00', 18000.00, 'Confirmed'),
(4, '2024-05-04 13:00:00', 18000.00, 'Confirmed'),
(5, '2024-05-05 14:00:00', 10000.00, 'Confirmed'),
(6, '2024-05-06 15:00:00', 10000.00, 'Confirmed'),
(7, '2024-05-07 16:00:00', 16000.00, 'Confirmed'),
(8, '2024-05-08 17:00:00', 16000.00, 'Confirmed'),
(9, '2024-05-09 18:00:00', 9000.00, 'Confirmed'),
(50, '2024-05-20 17:00:00', 160000.00, 'Confirmed');

INSERT INTO BookingPassenger (booking_id, passenger_id) VALUES
(1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6), (7, 7), (8, 8), (9, 9)
(11, 11), (12, 12), (13, 13), (14, 14), (15, 15), (16, 16), (17, 17), (18, 18), (19, 19), (20, 20);

INSERT INTO Ticket (ticket_id, booking_id, passenger_id, flight_schedule_id, ticket_number, ticket_price,
class_type, ticket_status) VALUES
(1, 1, 1, 1, 'TKT-1000001', 15000.00, 'Economy', 'Active'),
(2, 2, 2, 2, 'TKT-1000002', 15000.00, 'Economy', 'Active'),
(3, 3, 3, 3, 'TKT-1000003', 18000.00, 'Economy', 'Active'),
(4, 4, 4, 4, 'TKT-1000004', 18000.00, 'Economy', 'Active'),
(5, 5, 5, 5, 'TKT-1000005', 10000.00, 'Economy', 'Active'),
(6, 6, 6, 6, 'TKT-1000006', 10000.00, 'Economy', 'Active'),
(7, 7, 7, 7, 'TKT-1000007', 16000.00, 'Economy', 'Active'),
(8, 8, 8, 8, 'TKT-1000008', 16000.00, 'Economy', 'Active'),
(9, 9, 9, 9, 'TKT-1000009', 9000.00, 'Economy', 'Active'),
(10, 10, 10, 10, 'TKT-1000010', 9000.00, 'Economy', 'Active'),
(50, 50, 20, 50, 'TKT-1000050', 160000.00, 'First Class', 'Active');

INSERT INTO CheckIn (check_in_id, ticket_id, check_in_datetime, check_in_method, counter_number) VALUES
(1, 1, '2024-06-01 05:00:00', 'Online', NULL),
(2, 2, '2024-06-01 07:30:00', 'Counter', 'C12'),
(3, 3, '2024-06-01 11:00:00', 'Kiosk', NULL),
(4, 4, '2024-06-02 06:00:00', 'Online', NULL),
(5, 5, '2024-06-02 12:00:00', 'Counter', 'C05'),
(6, 6, '2024-06-03 04:00:00', 'Online', NULL),
(7, 7, '2024-06-03 15:00:00', 'Kiosk', NULL),
(8, 8, '2024-06-04 08:00:00', 'Counter', 'C08'),
(9, 9, '2024-06-04 13:00:00', 'Online', NULL),
,(50, 50, '2024-06-25 06:00:00', 'Counter', 'C25');
```

```sql
INSERT INTO Boarding (boarding_id, ticket_id, boarding_datetime, boarding_status) VALUES
(1, 1, '2024-06-01 07:20:00', 'Boarded'),
(2, 2, '2024-06-01 09:50:00', 'Boarded'),
(3, 3, '2024-06-01 13:20:00', 'Boarded'),
(4, 4, '2024-06-02 08:20:00', 'Boarded'),
(5, 5, '2024-06-02 14:20:00', 'Boarded'),
(6, 6, '2024-06-03 06:20:00', 'Boarded'),
(7, 7, '2024-06-03 17:20:00', 'Boarded'),
(8, 8, '2024-06-04 10:20:00', 'Boarded'),
(9, 9, '2024-06-04 15:20:00', 'Boarded'),
(10, 10, '2024-06-05 07:50:00', 'Boarded'),
(50, 50, '2024-06-25 08:20:00', 'Boarded');

INSERT INTO Connections (connection_id, connection_name, connection_type, total_layover_duration_min)
VALUES
(1, 'KHI-LHE-ISB', 'Domestic', 120),
(2, 'ISB-LHE-KHI', 'Domestic', 90),
(3, 'KHI-DXB-LHR', 'International', 240),
(4, 'LHR-DXB-KHI', 'International', 180),
(5, 'LHE-DOH-JFK', 'International', 300),
(6, 'JFK-DOH-LHE', 'International', 210),
(7, 'ISB-IST-YYZ', 'International', 360),
(8, 'YYZ-IST-ISB', 'International', 240),
(9, 'PEW-DXB-JED', 'International', 150),
(20, 'SYD-PEK-ISB', 'International', 400);

INSERT INTO ConnectionSegment (connection_segment_id, connection_id, flight_schedule_id, segment_order)
VALUES
(1, 1, 1, 1), (2, 1, 5, 2), (3, 2, 4, 1), (4, 2, 2, 2), (5, 3, 17, 1),
(6, 3, 27, 2), (7, 4, 28, 1), (8, 4, 18, 2), (9, 5, 35, 1), (10, 5, 31, 2),
(11, 6, 32, 1), (12, 6, 36, 2), (13, 7, 45, 1), (14, 7, 33, 2
;

INSERT INTO MaintenanceType (maintenance_type_id, type_name, type_description) VALUES
(1, 'A Check', 'Light maintenance, every 400-600 hours'),
(2, 'B Check', 'Light to medium maintenance, every 6-8 months'),
(3, 'C Check', 'Heavy maintenance, every 20-24 months'),
(4, 'D Check', 'Extensive structural heavy check, every 6-10 years'),
(5, 'Line Maintenance', 'Pre-flight, transit, and daily checks'),
(6, 'Engine Overhaul', 'Complete teardown and rebuild of engine'),
(19, 'Fire Suppression Test', 'Testing cargo and engine fire systems'),
(20, 'Bird Strike Inspection', 'Inspection following reported bird strike');

INSERT INTO MaintenanceEvent (maintenance_event_id, aircraft_id, maintenance_type_id, event_date,
completion_date, event_description, status) VALUES
(1, 1, 5, '2024-05-01', '2024-05-01', 'Daily transit check', 'Completed'),
(2, 2, 5, '2024-05-02', '2024-05-02', 'Daily transit check', 'Completed'),
(3, 3, 1, '2024-05-03', '2024-05-04', 'A Check schedule', 'Completed'),
(4, 4, 2, '2024-05-05', '2024-05-07', 'B Check routine', 'Completed'),
(6, 6, 12, '2024-05-08', '2024-05-08', 'Brake pad inspection', 'Completed'),
(7, 7, 5, '2024-05-10', '2024-05-10', 'Daily check', 'Completed'),
(8, 8, 16, '2024-05-11', '2024-05-11', 'FMC software patch', 'Completed'),
(10, 10, 10, '2024-05-16', '2024-05-17', 'APU routine service', 'Completed'),
(11, 11, 5, '2024-05-18', '2024-05-18', 'Line check', 'Completed'),
(50, 11, 5, '2024-06-30', NULL, 'Transit ongoing', 'Scheduled');
```

```sql
INSERT INTO MaintenanceAssignment (maintenance_assignment_id, employee_id, maintenance_event_id,
assigned_date) VALUES
(1, 9, 1, '2024-05-01'),
(2, 10, 2, '2024-05-02'),
(3, 8, 3, '2024-05-03'),
(4, 9, 4, '2024-05-05'),
(5, 10, 5, '2024-05-06'),
(6, 8, 6, '2024-05-08'),
(7, 9, 7, '2024-05-10'),
(50, 8, 50, '2024-06-30');

INSERT INTO FlightCrewAssignment (flight_crew_assignment_id, employee_id, flight_schedule_id, crew_role,
assigned_date) VALUES
(1, 2, 1, 'Captain', '2024-05-25'),
(2, 4, 1, 'First Officer', '2024-05-25'),
(3, 3, 1, 'Lead Cabin Crew', '2024-05-25'),
(4, 2, 2, 'Captain', '2024-05-25'),
(5, 4, 2, 'First Officer', '2024-05-25'),
(6, 3, 2, 'Lead Cabin Crew', '2024-05-25'),
(7, 11, 3, 'Captain', '2024-05-26'),
(8, 6, 3, 'First Officer', '2024-05-26'),
(9, 7, 3, 'Flight Attendant', '2024-05-26'),
(50, 21, 17, 'First Officer', '2024-06-02');

INSERT INTO SeatAssignment (seat_assignment_id, seat_id, ticket_id, flight_schedule_id,
assignment_datetime) VALUES
(1, 1, 1, 1, '2024-06-01 05:05:00'),
(2, 6, 2, 2, '2024-06-01 07:35:00'),
(3, 11, 3, 3, '2024-06-01 11:05:00'),
(4, 2, 4, 4, '2024-06-02 06:05:00'),
(5, 31, 5, 5, '2024-06-02 12:05:00'),
(6, 36, 6, 6, '2024-06-03 04:05:00'),
(7, 41, 7, 7, '2024-06-03 15:05:00'),
(8, 46, 8, 8, '2024-06-04 08:05:00'),
(9, 7, 9, 9, '2024-06-04 13:05:00'),
(50, 13, 50, 50, '2024-06-25 06:05:00');

INSERT INTO Baggage (baggage_id, check_in_id, weight_kg, baggage_type, baggage_tag_number,
baggage_status) VALUES
(1, 1, 20.50, 'Checked', 'BAG1001', 'Loaded'),
(2, 2, 18.00, 'Checked', 'BAG1002', 'Loaded'),
(3, 3, 22.10, 'Checked', 'BAG1003', 'Loaded'),
(4, 4, 15.40, 'Checked', 'BAG1004', 'Loaded'),
(5, 5, 23.00, 'Checked', 'BAG1005', 'Loaded'),
(6, 6, 19.50, 'Checked', 'BAG1006', 'Loaded'),
(7, 7, 21.00, 'Checked', 'BAG1007', 'Loaded'),
(8, 8, 14.50, 'Checked', 'BAG1008', 'Loaded'),
(9, 9, 25.00, 'Checked', 'BAG1009', 'Loaded'
(50, 50, 22.50, 'Checked', 'BAG1050', 'Loaded');

INSERT INTO Payment (payment_id, booking_id, payment_datetime, payment_amount, payment_method,
payment_status, transaction_reference) VALUES
(1, 1, '2024-05-01 10:05:00', 15000.00, 'Credit Card', 'Completed', 'TXN-0001'),
(2, 2, '2024-05-02 11:05:00', 15000.00, 'Debit Card', 'Completed', 'TXN-0002'),
```

```sql
(3, 3, '2024-05-03 12:05:00', 18000.00, 'Bank Transfer', 'Completed', 'TXN-0003'),
(4, 4, '2024-05-04 13:05:00', 18000.00, 'Credit Card', 'Completed', 'TXN-0004'),
(5, 5, '2024-05-05 14:05:00', 10000.00, 'JazzCash', 'Completed', 'TXN-0005'), (6, 6, '2024-05-06 15:05:00',
10000.00, 'EasyPaisa', 'Completed', 'TXN-0006'),
(7, 7, '2024-05-07 16:05:00', 16000.00, 'Credit Card', 'Completed', 'TXN-0007'),
(8, 8, '2024-05-08 17:05:00', 16000.00, 'Debit Card', 'Completed', 'TXN-0008'),
(9, 9, '2024-05-09 18:05:00', 9000.00, 'Credit Card', 'Completed', 'TXN-0009'),
(50, 50, '2024-05-20 17:05:00', 160000.00, 'Credit Card', 'Completed', 'TXN-0050');

INSERT INTO Invoice (invoice_id, payment_id, invoice_date, invoice_amount, billing_address, invoice_status)
VALUES
(1, 1, '2024-05-01', 15000.00, 'Clifton, Karachi', 'Paid'),
(2, 2, '2024-05-02', 15000.00, 'DHA, Lahore', 'Paid'),
(3, 3, '2024-05-03', 18000.00, 'F-8, Islamabad', 'Paid'),
(4, 4, '2024-05-04', 18000.00, 'Gulberg, Lahore', 'Paid'),
(5, 5, '2024-05-05', 10000.00, 'PECHS, Karachi', 'Paid'),
(6, 6, '2024-05-06', 10000.00, 'Blue Area, Islamabad', 'Paid'),
(7, 7, '2024-05-07', 16000.00, 'Saddar, Karachi', 'Paid'),
(8, 8, '2024-05-08', 16000.00, 'Model Town, Lahore', 'Paid'),
(9, 9, '2024-05-09', 9000.00, 'G-11, Islamabad', 'Paid'), (10, 10, '2024-05-
(50, 50, '2024-05-20', 160000.00, 'F-7, Islamabad', 'Paid');
```

7. 20 Reporting Queries

```sql
-- Query 1: [Shows all active planes in the airline]
SELECT
   registration_number,
   manufacturer,
   model,
   total_seats
FROM Aircraft
WHERE is_active = 1;

-- Query 2: [Finds all Pakistani passengers born after 1990]
SELECT
   first_name,
   last_name,
   passport_number,
   date_of_birth
FROM Passenger
WHERE nationality = 'Pakistani'
  AND date_of_birth > '1990-12-31';

-- Query 3: [Counts the total number of confirmed bookings]
SELECT
   COUNT(booking_id) AS total_confirmed_bookings
FROM Booking
WHERE booking_status = 'Confirmed';

-- Query 4: [Shows the top 5 most expensive flights]
SELECT TOP 5
   flight_number,
   airline_name,
   base_price
FROM Flight
ORDER BY base_price DESC;
-- Query 5: [Shows flight routes and changes distance from kilometers to miles]
SELECT
   route_id,
   distance_km,
   (distance_km * 0.621371) AS distance_miles,
   estimated_duration_min
FROM Route;
-- Query 6: [Shows employee names and their job titles]
SELECT
   e.first_name,
   e.last_name,
   r.role_name
FROM Employee e
JOIN EmployeeRole er ON e.employee_id = er.employee_id
JOIN Role r ON er.role_id = r.role_id;

-- Query 7: [Counts how many flights each airline has]
SELECT
   airline_name,
   COUNT(flight_id) AS total_flights
FROM Flight
GROUP BY airline_name
```

```sql
-- Query 8: [Finds airports that have more than 3 gates]
SELECT
   a.airport_name,
   COUNT(g.gate_id) AS total_gates
FROM Airport a
JOIN Gate g ON a.airport_id = g.airport_id
GROUP BY a.airport_name
HAVING COUNT(g.gate_id) > 3;


-- Query 9: [Shows employees and the names of their managers]
SELECT
   emp.first_name AS employee_name,
   emp.last_name AS employee_surname,
   mgr.first_name AS manager_name,
   mgr.last_name AS manager_surname
FROM Employee emp
LEFT JOIN Employee mgr ON emp.manager_id = mgr.employee_id;


-- Query 10: [Shows plane maintenance tasks done in May 2024]
SELECT
   ac.registration_number,
   mt.type_name,
   me.event_date,
   me.status
FROM MaintenanceEvent me
JOIN Aircraft ac ON me.aircraft_id = ac.aircraft_id
JOIN MaintenanceType mt ON me.maintenance_type_id = mt.maintenance_type_id
WHERE me.event_date BETWEEN '2024-05-01' AND '2024-05-31';


-- Query 11: [Finds passengers who spent more than 100,000 on a booking]
SELECT
   first_name,
   last_name,
   email
FROM Passenger
WHERE passenger_id IN (
   SELECT passenger_id
   FROM BookingPassenger bp
   JOIN Booking b ON bp.booking_id = b.booking_id
   WHERE b.total_amount > 100000
);


-- Query 12: [Finds flights that cost more than the average flight price]
SELECT
   flight_number,
   airline_name,
   base_price
FROM Flight f
WHERE base_price > (
   SELECT AVG(base_price) FROM Flight
);
```

```sql
-- Query 13: [Calculates total money made from tickets for each scheduled flight]
SELECT
  fs.flight_schedule_id,
  f.flight_number,
  SUM(t.ticket_price) AS total_revenue,
  COUNT(t.ticket_id) AS tickets_sold
FROM FlightSchedule fs
JOIN Flight f ON fs.flight_id = f.flight_id
JOIN Ticket t ON fs.flight_schedule_id = t.flight_schedule_id
GROUP BY fs.flight_schedule_id, f.flight_number;

-- Query 14: [Makes a list of passengers, their flight numbers, and seat numbers]
SELECT
  p.first_name,
  p.last_name,
  f.flight_number,
  fs.scheduled_departure,
  s.seat_number,
  t.class_type
FROM Ticket t
JOIN Passenger p ON t.passenger_id = p.passenger_id
JOIN FlightSchedule fs ON t.flight_schedule_id = fs.flight_schedule_id
JOIN Flight f ON fs.flight_id = f.flight_id
JOIN SeatAssignment sa ON t.ticket_id = sa.ticket_id
JOIN Seat s ON sa.seat_id = s.seat_id;

-- Query 15: [Finds registered passengers who have never booked a ticket]
SELECT
  first_name,
  last_name,
  email
FROM Passenger
WHERE passenger_id NOT IN (
  SELECT passenger_id FROM BookingPassenger
);

-- Query 16: [Checks if a flight left on time or was delayed]
SELECT
  f.flight_number,
  fs.scheduled_departure,
  fs.actual_departure,
  CASE
    WHEN fs.actual_departure > fs.scheduled_departure THEN 'Delayed'
    WHEN fs.actual_departure IS NULL THEN 'Pending'
    ELSE 'On Time'
  END AS departure_status
FROM FlightSchedule fs
JOIN Flight f ON fs.flight_id = f.flight_id;


-- Query 17: [Calculates the total weight of all bags on each flight]
SELECT
  fs.flight_schedule_id,
  f.flight_number,
```

```sql
    SUM(b.weight_kg) AS total_baggage_weight
FROM FlightSchedule fs
JOIN Flight f ON fs.flight_id = f.flight_id  -- This was the missing line!
JOIN Ticket t ON fs.flight_schedule_id = t.flight_schedule_id
JOIN CheckIn c ON t.ticket_id = c.ticket_id
JOIN Baggage b ON c.check_in_id = b.check_in_id
GROUP BY fs.flight_schedule_id, f.flight_number
ORDER BY total_baggage_weight DESC;

-- Query 18: [Finds the most used check-in method for flights leaving from Karachi]
SELECT check_in_method, COUNT(*) AS usage_count
FROM CheckIn
WHERE ticket_id IN (
    SELECT t.ticket_id
    FROM Ticket t
    JOIN FlightSchedule fs ON t.flight_schedule_id = fs.flight_schedule_id
    JOIN Flight f ON fs.flight_id = f.flight_id
    JOIN Route r ON f.route_id = r.route_id
    JOIN Airport a ON r.origin_airport_id = a.airport_id
    WHERE a.iata_code = 'KHI'
)
GROUP BY check_in_method
ORDER BY usage_count DESC;


-- Query 19: [Counts how many passengers have traveled between two specific cities]
SELECT
    orig.city AS origin,
    dest.city AS destination,
    COUNT(t.passenger_id) AS total_passengers
FROM Route r
JOIN Airport orig ON r.origin_airport_id = orig.airport_id
JOIN Airport dest ON r.destination_airport_id = dest.airport_id
JOIN Flight f ON r.route_id = f.route_id
JOIN FlightSchedule fs ON f.flight_id = fs.flight_id
JOIN Ticket t ON fs.flight_schedule_id = t.flight_schedule_id
GROUP BY orig.city, dest.city
ORDER BY total_passengers DESC;


-- Query 20: [Shows total money collected by each payment method, ignoring methods under 50,000]
SELECT
    payment_method,
    COUNT(payment_id) AS number_of_transactions,
    SUM(payment_amount) AS total_revenue_collected
FROM Payment
WHERE payment_status = 'Completed'
GROUP BY payment_method
HAVING SUM(payment_amount) >= 50000
ORDER BY total_revenue_collected DESC;
```